



Optional: Upload the store transactions dataset to BigQuery

In the next video, the instructor uses a specific dataset. The instructions in this reading are provided for you to upload the same dataset in your BigQuery console so you can follow along.

You must have a BigQuery account to follow along.

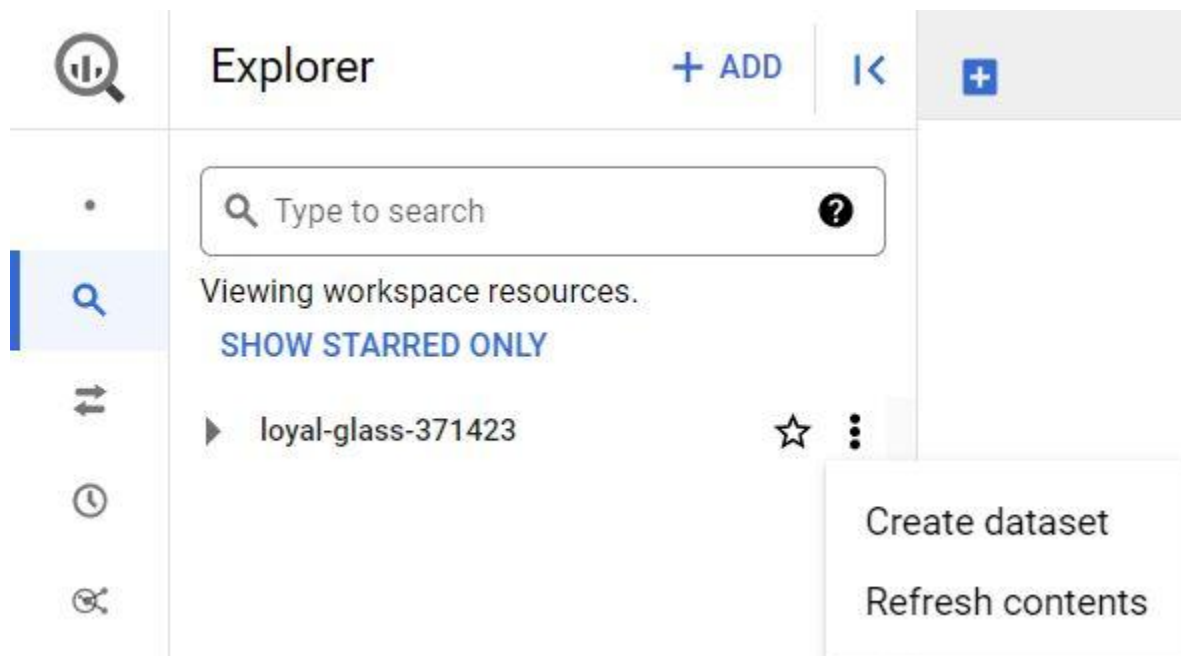
Prepare for the next video

- First, download the .csv file from the attachment below.
- Next, complete the steps below in your BigQuery console to upload the Store Transaction dataset.

Note: These steps will be different from what you performed before. In previous instances, you selected the Auto detect check box to allow BigQuery to auto-detect the schema. This time, you will choose to create the schema by editing it as text. This method can be used when BigQuery doesn't automatically set the desired type for a particular field. In this case, you will specify **STRING** instead of **FLOAT** as the type for the purchase_price field.

Step 1: Open your BigQuery console and select the expansion arrow next to the project you'll upload the data to. Examine the datasets listed under your project. If a customer_data dataset is listed, go to step 5; otherwise, continue with step 2.

Step 2: In the Explorer on the left, click the Actions icon (three vertical dots) next to your project name and select Create dataset.



Step 3: In the Create dataset window, enter `customer_data` for the Dataset ID. Make sure the *Location type* is set to Multi-region, US (Multiple regions in United States), and all the default *Advanced options* remain set to the Google-managed encryption key option.

Create dataset

Project ID
loyal-glass-371423

CHANGE

Dataset ID *

customer_data

Letters, numbers, and underscores allowed

Location type ?

☐ Region
Specify a region to colocate your datasets with other GCP services.

☒ Multi-region
Allow BigQuery to select a region within a group to achieve higher quota limits.

Multi-region *

US (multiple regions in United States)

Default table expiration

☐ Enable table expiration ?

Default maximum table age Days

Advanced options

CREATE DATASET

CANCEL

Step 4: Click CREATE DATASET (blue button) to add the dataset to your project.

Step 5: In the Explorer pane, click on the expansion arrow under your project name, and then click the customer_data dataset.

Explorer + ADD K

customer_data

CREATE TABLE

Dataset info

Dataset ID	loyal-glass-371423.customer_data
Created	Jul 5, 2023, 11:45:41 AM UTC-5
Default table expiration	60 days
Last modified	Jul 5, 2023, 11:45:41 AM UTC-5
Data location	US
Description	
Default collation	
Default rounding mode	ROUNDING_MODE_UNSPECIFIED
Time travel window	7 days
Case insensitive	false
Labels	
Tags	

Step 6: In the Dataset info page, click the blue + CREATE TABLE button to open the Create table window.

Step 7: Under Source, for the Create table from selection, choose where the data will be coming from.

- Select Upload.
- Click Browse to select the Store Transaction Table .csv file you downloaded.
- Choose CSV from the file format drop-down.

Create table

Source

Create table from
Upload

Select file *
Lauren-s-Furniture-Store-Transaction.csv

File format
CSV

Destination

Project *
loyal-glass-371423

Dataset *
customer_data

Table *
customer_purchase

Unicode letters, marks, numbers, connectors, dashes or spaces allowed.

Table type
Native table

Schema

☐ Auto detect

☒ Edit as text

Step 8: For Table name, enter customer_purchase.

Step 9: For Schema, click the toggle switch for Edit as text. This opens up a box for the text.

Step 10: Copy and paste the following text into the box. Be sure to include the opening and closing brackets. They are required.

```
[
  {
    "description": "date",
    "mode": "NULLABLE",
    "name": "date",
    "type": "DATETIME"
  },
  {
    "description": "transaction id",
    "mode": "NULLABLE",
    "name": "transaction_id",
    "type": "INTEGER"
  },
  {
    "description": "customer id",
    "mode": "NULLABLE",
    "name": "customer_id",
    "type": "INTEGER"
  },
  {
    "description": "product name",
    "mode": "NULLABLE",
    "name": "product",
    "type": "STRING"
  },
  {
    "description": "product_code",
    "mode": "NULLABLE",
    "name": "product_code",
    "type": "STRING"
  },
  {
    "description": "product color",
    "mode": "NULLABLE",
```

```
"name": "product_color",
"type": "STRING"
},
{
  "description": "product price",
  "mode": "NULLABLE",
  "name": "product_price",
  "type": "FLOAT"
},
{
  "description": "quantity purchased",
  "mode": "NULLABLE",
  "name": "purchase_size",
  "type": "INTEGER"
},
{
  "description": "purchase price",
  "mode": "NULLABLE",
  "name": "purchase_price",
  "type": "STRING"
},
{
  "description": "revenue",
  "mode": "NULLABLE",
  "name": "revenue",
  "type": "FLOAT"
}
]
```

Step 11: Scroll down and expand the Advanced options section.

Step 12: For the Header rows to skip field, enter 1.

NOTE: It is very important that you don't skip the last step, or you will receive 'parsing' errors, as BigQuery will try to apply the schema editing functions to the title row.

Partition and cluster settings

Partitioning
No partitioning

Clustering order

Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options

Write preference
Write if empty

Number of errors allowed
0

☐ Unknown values

Field delimiter
Comma

Quote character
Double quote

Header rows to skip
1

CREATE TABLE CANCEL

Step 13: Click Create table (blue button). You will now see the customer_purchase table under your customer_data dataset in your Explorer pane.

Step 14: Click the customer_purchase table and in the Schema tab, confirm that the schema matches the schema shown below.

[SCHEMA](#)[DETAILS](#)[PREVIEW](#)

Table schema

 **Filter** Enter property name or value

Field name	Type	Mode	Policy Tags 	Description
date	DATETIME	NULLABLE		date
transaction_id	INTEGER	NULLABLE		transaction id
customer_id	INTEGER	NULLABLE		customer id
product	STRING	NULLABLE		product name
product_code	STRING	NULLABLE		product_code
product_color	STRING	NULLABLE		product color
product_price	FLOAT	NULLABLE		product price
purchase_size	INTEGER	NULLABLE		quantity purchased
purchase_price	STRING	NULLABLE		purchase price
revenue	FLOAT	NULLABLE		revenue

Step 15: Click the Preview tab and confirm that your data matches the data shown below.

SCHEMA		DETAILS		PREVIEW						
Row	date	transaction_id	customer_id	product	product_code	product_color	product_price	purchase_size	purchase_price	revenue
1	2020-08-29T00:00:00	9900	1980	fan	SKU83503	brass	13.99	2	27.98	27.98
2	2020-05-01T00:00:00	12315	2463	fan	SKU83503	brass	13.99	2	27.98	27.98
3	2020-12-12T00:00:00	9890	1978	fan	SKU83503	white	13.99	1	13.99	13.99
4	2020-02-16T00:00:00	46915	9383	fan	SKU83503	black	13.99	1	13.99	13.99
5	2020-12-28T00:00:00	44700	8940	fan	SKU83503	brass	13.99	2	27.98	27.98
6	2020-12-28T00:00:00	44700	8940	lamp	SKU95363	brass	45.99	5	160.965	229.95
7	2020-10-11T00:00:00	12560	2512	bed	SKU74796	brown	799.99	1	799.99	799.99
8	2020-06-29T00:00:00	9640	1928	couch	SKU31871	grey	1000.0	1	1000	1000.0
9	2020-11-10T00:00:00	22620	4524	couch	SKU31871	grey	1000.0	1	1000	1000.0
10	2020-02-05T00:00:00	49430	9886	couch	SKU31871	grey	1000.0	1	1000	1000.0
11	2020-04-28T00:00:00	49075	9815	couch	SKU31871	blue	1000.0	1	1000	1000.0
12	2020-01-23T00:00:00	37855	7571	couch	SKU31871	purple	1000.0	1	1000	1000.0
13	2020-02-16T00:00:00	9900	1980	couch	SKU31871	white	1000.0	1	1000	1000.0
14	2020-10-12T00:00:00	1675	335	couch	SKU31871	white	1000.0	1	1000	1000.0
15	2020-07-08T00:00:00	23435	4687	couch	SKU31871	brown	1000.0	1	1000	1000.0
16	2020-10-21T00:00:00	19105	3821	couch	SKU31871	black	1000.0	1	1000	1000.0
17	2020-12-30T00:00:00	48035	9607	rug	SKU19092	beige	89.85	3	269.55	269.55
18	2020-09-25T00:00:00	31615	6323	rug	SKU19092	grey	89.85	2	179.7	179.7
19	2020-08-11T00:00:00	31775	6355	rug	SKU19092	grey	89.85	1	89.85	89.85
20	2020-09-07T00:00:00	21870	4374	rug	SKU19092	beige	89.85	3	269.55	269.55
21	2020-07-22T00:00:00	45400	9080	vase	SKU99037	green	9.99	2	19.98	19.98
22	2020-07-22T00:00:00	45400	9080	mirror	SKU97721	brass	199.95	1	199.95	199.95
23	2020-07-22T00:00:00	45400	9080	bookcase	SKU15757	brown	58.89	1	58.89	58.89
24	2020-04-29T00:00:00	26530	5306	desk	SKU40294	brown	169.95	1	169.95	169.95
25	2020-03-22T00:00:00	14080	2816	desk	SKU40294	brown	169.95	1	169.95	169.95
26	2020-06-04T00:00:00	6340	1268	desk	SKU40294	white	169.95	1	169.95	169.95
27	2020-02-28T00:00:00	21485	4297	chair	SKU99664	black	234.5	1	234.5	234.5
28	2020-06-06T00:00:00	24785	4957	<i>null</i>	SKU77417	beige	399.95	1	399.95	399.95
29	2020-06-06T00:00:00	24785	4957	<i>null</i>	SKU13351	grey	299.99	1	299.99	299.99

Congratulations, you are now ready to follow along with the video!

Key Concept:

- **CSV File:** A CSV (Comma-Separated Values) file is a common data format used for storing tabular data, which can be easily imported into BigQuery.
- **Data Types:** Data types specify the kind of data that can be stored in a field, such as STRING, INTEGER, or FLOAT, which is essential for accurate data processing.
- **Schema:** The schema defines the structure of a table, including the names and data types of its fields, which is crucial for data integrity.
- **BigQuery:** BigQuery is a fully-managed, serverless data warehouse that enables scalable analysis over large datasets.
- **Dataset:** A dataset in BigQuery is a collection of tables that allows users to organize and manage their data effectively.